

steps for java installation:

Step 1: Install Java (JDK)

Ubuntu usually comes with OpenJDK in its repositories.

Open a terminal and run:

```
sudo apt update
```

```
sudo apt install
```

Check Java installation:

```
java -version
```

You should see something like:

```
openjdk version "11.0.x" ...
```

Step 2: Install MySQL Server

You can install MySQL to test JDBC with a real database:

```
sudo apt install mysql-server
```

Secure the installation:

Start MySQL service:

```
sudo systemctl start mysql
```

Log into MySQL shell:

```
sudo mysql -u root -p
```

Create a test database and user:

```
mysql> CREATE DATABASE testdb;
```

```
mysql> USE testdb;
```

```
mysql> CREATE TABLE users (
```

```
-> id INT PRIMARY KEY AUTO_INCREMENT,
```

```
-> name VARCHAR(50),
```

```
-> email VARCHAR(50)
```

```
-> );
```

```
mysql> INSERT INTO users (name, email) VALUES ('Alice', 'alice@example.com'), ('Bob',  
'bob@example.com');
```

```
mysql> CREATE USER 'testuser'@'localhost' IDENTIFIED BY 'testpass';
```

```
mysql> GRANT ALL PRIVILEGES ON testdb.* TO 'testuser'@'localhost';
```

```
mysql> FLUSH PRIVILEGES;
```

```
mysql> EXIT;
```